

Hanyu Da

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EDUCATION

New York University GPA: 3.8/4.0

New York, NY

M.S. in Computer Engineering

Sept 2025 – May 2027

Coursework: Efficient AI & Hardware Accelerator Design, Internet Architecture & Protocols, Machine Learning, System Optimization, Embedded Systems, Cloud Computing, High-Speed Networks

New York University

New York, NY

B.S. in Applied Mathematics, Minor in Computer Science

Sept 2021 – May 2025

Coursework: Machine Learning, Data Structures & Algorithms, Computer Networks, Mathematical Modeling, Statistics & Probability, Object-Oriented Programming, Numerical Analysis

SKILLS & INTERESTS

Technical: Python (Pandas, NumPy, PyTorch, SciPy, Scikit-learn), C++, SQL, Java, JavaScript, React.js, Django, Swift, HTML/CSS, PHP, FAISS, Infomap, UNIX, Power BI, Excel

Machine Learning: Deep Learning, CNNs, Clustering, SVM, Logistic Regression, Random Forest, PCA

AI Systems: Model Compression, Structured Pruning, FP16 Mixed-Precision Inference, RAG, FAISS Retrieval, LLM Local Deployment & Fine-tuning (Qwen)

Languages: English (fluent), Mandarin (native)

PROFESSIONAL EXPERIENCE

WeatherRemover: Model Compression & Efficient Inference

New York, NY

Group Project Leader

Feb 2026 – May 2026

- Profiled the baseline CMFormer model on an NVIDIA RTX 3090 using PyTorch profiler to identify computational hotspots and memory bottlenecks across attention layers and convolution blocks, establishing a performance baseline for targeted compression.
- Applied structured pruning and FP16 mixed-precision inference to compress the model; built reproducible evaluation scripts benchmarking PSNR, SSIM, latency, throughput, GPU memory, parameter count, MACs, and FLOPs across baseline and compressed variants.
- Achieved the best accuracy-efficiency tradeoff with Prune-refine4 + FP16, reducing parameters by ~30% and inference latency by ~39% while incurring only ~0.41 dB PSNR and ~0.0033 SSIM degradation.

Chinasoft International Technology Service Ltd.

Nanjing, China

Machine Learning Engineer Intern

May 2025 – Aug 2025

- Built a modular face-clustering pipeline integrating Inception-ResNet-v2 and HyNet for deep feature extraction, enabling identity-based grouping of 100K+ unlabeled face images.
- Accelerated similarity search with FAISS-based KNN and cosine-distance graph construction, reducing search time by 80% and enabling 5× dataset scalability.
- Developed a noise-filtering and edge-linking algorithm to refine graph topology, improving clustering precision by 15%; built a CLI evaluation framework supporting FMI, B-cubed F1, and NMI for automated benchmarking across experiments.

NYU LLM Course Advising Assistant

New York, NY

Group Project Leader

Oct 2025 – Dec 2025

- Built a Flask web application integrating a locally deployed Qwen LLM with an end-to-end RAG pipeline—ingesting syllabus PDFs, generating E5 embeddings, and indexing/retrieving via FAISS.
- Improved retrieval relevance through query rewriting, course-code boosting, exam/grading keyword boosting, and dominant-course filtering; exposed core API endpoints with an interactive chat UI.
- Implemented memory-aware advising features: persistent conversational memory store, course selection and comparison logic, and automatic weekly schedule generation with HTML timetable rendering, conflict reporting, and pytest unit and integration tests.

Innovation AI

New York, NY

Data Analyst Intern

June 2024 – Aug 2024

- Cleaned and validated movie performance datasets for CVT prediction—imputing 5% missing values, preparing categorical/numerical features, and examining correlations among box office revenue, IMDb ratings, awards, and release-year trends.
- Investigated the impact of genres on numerical performance metrics and assessed how awards influence box office revenue, IMDb ratings, and daily CVT.
- Built and compared Linear Regression, Lasso, and Random Forest models on train/test splits; identified Random Forest as the best-performing CVT predictor.